

Backend Developer with production experience building high-concurrency, fault-tolerant systems for booking- and payment-critical workloads. Specializes in distributed concurrency control (Redis SET NX locking, Lua-scripted token-bucket rate limiting), ACID-compliant transaction design (PostgreSQL, GORM), and cloud-native microservices deployed via gRPC on AWS EKS. Delivered a venue-booking engine that eliminates double-booking under concurrent traffic and a double-entry wallet ledger guaranteeing credit-debit consistency across async payment webhooks. Strong foundation in DSA, HLD/LLD, and translating complex business logic into clean, testable, cost-efficient architectures.

SKILLS

Language & Core	Go (Golang), Goroutines & Channels, Sync Primitives (Mutex, WaitGroup), Context & Cancellation, Memory Profiling (pprof), SQL, Bash
Backend	Gin, REST APIs, gRPC & Protocol Buffers, API Gateway & Routing, Token Bucket Rate Limiting, Service-to-Service Communication
Databases & Cache	PostgreSQL (ACID, Indexing, Row Locking, Txns via GORM), Redis (Distributed Locking, Lua Scripting, Caching), MongoDB (Aggregation), Distributed Concurrency Control
Message Queues	RabbitMQ, Transactional Outbox Pattern, Async Event Publishing
API & Security	RESTful Design, RBAC, JWT, Access & Refresh Tokens, Webhooks, Google OAuth 2.0, HMAC-SHA256 Signature Verification
Cloud & DevOps	Linux (WSL/Ubuntu/Alpine), Docker (Multi-Stage Builds), Kubernetes (AWS EKS), AWS (EC2, S3 Presigned URLs, IAM, Lambda), NGINX, GitHub Actions, CI/CD
Architecture	Clean Architecture, SOLID Principles, Double-Entry Ledger Design, Monolith to Microservices Refactoring, Event-Driven Architecture
Integrations	RapidAPI, Razorpay Payment Gateway, Cloudinary, Brevo
Testing & QA	Unit & Integration Testing (Go, testify), Table-Driven Testing, Postman, Swagger

PROJECTS

BookMyVenue — Distributed Venue Reservation & Transaction Engine



June 2026 — Present

- Engineered a 10-minute distributed inventory-hold engine using Redis SET NX with TTL and rollback compensation, eliminating double-booking collisions under high concurrent checkout traffic without ever locking the primary database.
- Designed a 2-stage (30% advance / 70% final) payment workflow integrated with Razorpay, guaranteeing financial integrity via HMAC-SHA256 signature verification and atomic 4-table PostgreSQL transactions with append-only audit logging for full traceability.
- Implemented an in-memory Token Bucket rate limiter in Redis using custom Lua scripts with fail-open safety, protecting public authentication and booking APIs against brute-force attacks and bot scrapers with sub-millisecond overhead.
- Built a direct-to-S3 presigned upload workflow that removes the backend entirely from the file-transfer path, eliminating server bandwidth bottlenecks for venue image uploads.
- Architected a shadow-copy draft system (VenueEditDraft) allowing venue owners to submit major edits for admin review without corrupting or unpublishing the live production listing.
- Designed atomic slot-replacement transaction bounds (gorm.Transaction) and prepared RabbitMQ event producers for async booking notifications, decoupling notification delivery from the booking critical path.

VogueLuxe — Cloud-Native Microservices E-Commerce Platform



Feb 2025 — Jun 2025

- Refactored a monolithic backend by extracting high-velocity domains (Wallet & Wishlist) into standalone gRPC microservices using Protocol Buffers, achieving strict fault isolation and sub-millisecond inter-service RPC latency.
- Built a reverse-proxy API Gateway handling centralized JWT authentication, role-based access control (RBAC), and protocol translation from incoming client HTTP requests to internal gRPC invocations.
- Designed a double-entry wallet ledger enforcing strict credit-debit balance consistency across asynchronous Razorpay payment webhooks, preventing fund mismatches during retries or out-of-order events.
- Containerized all services with multi-stage Docker builds to minimize image size, and automated CI/CD deployment via GitHub Actions into an AWS EKS Kubernetes cluster with zero-downtime rolling updates.

EDUCATION & TRAINING

Backend Engineering Program

Brototype Sep 2024 — May 2026

- Intensive project-based program covering Go backend engineering, advanced DSA, microservices architecture, Kubernetes deployments, and weekly technical code reviews.

Plus Two — Computer Commerce

Kerala Board of Higher Secondary Education, India Mar 2024